DGR LAB1 STP v1.5 - Jorge Parra ITIT 13104

Universidad La Salle Chihuahua

Diseño y Gestión de Redes

LAB 1 — Spanning Tree

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1. Introducción

> En esta práctica se implementó y verificó el funcionamiento de Spanning Tree Protocol (STP) en una topología jerárquica redundante. Se configuró Rapid-PVST en switches de acceso, distribución y núcleo, con los switches CORE como referencia para la elección del Root Bridge. También se analizaron los roles y estados de puertos para identificar enlaces en **Forwarding** y enlaces **bloqueados** para evitar bucles de Capa 2.

La verificación se realizó principalmente con `show spanning-tree vlan`, revisando modo, prioridades, roles y estados por VLAN.

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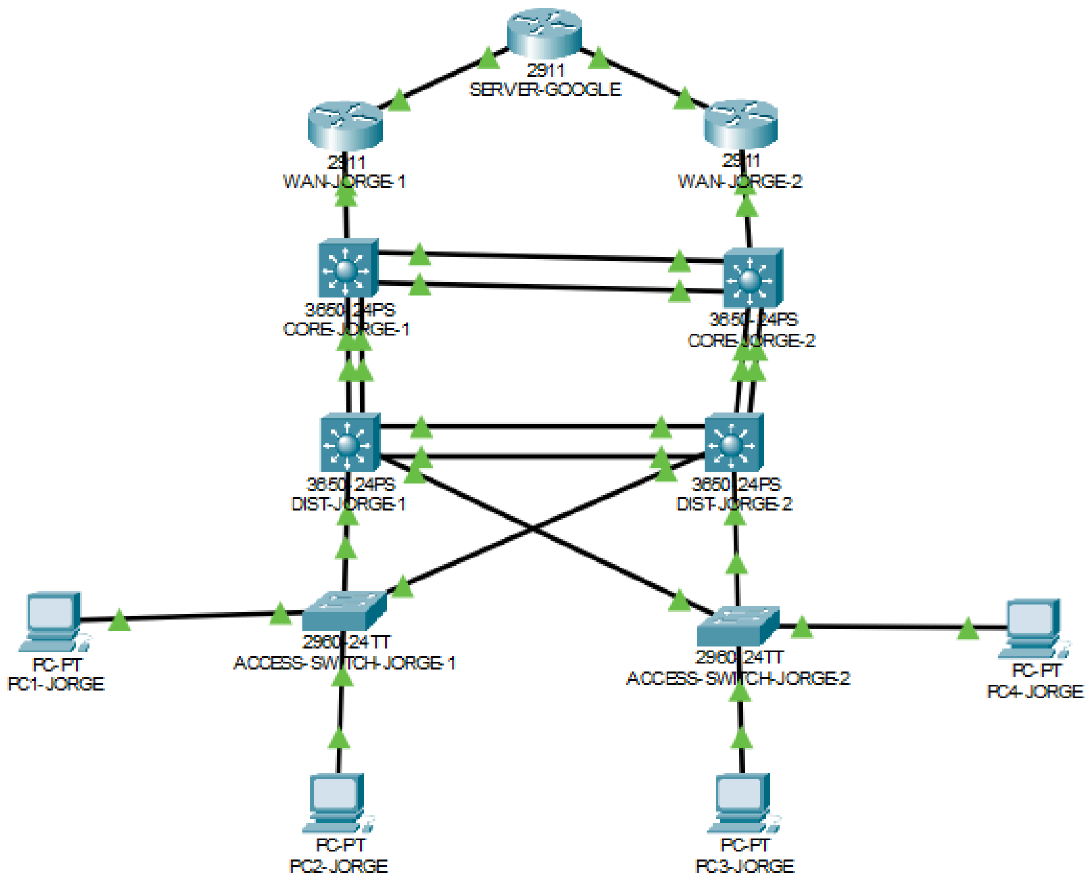
2. Evidencia y explicación

2.1 Topología



2.2 Capturas (comandos / resultados)

ACCESS-SWITCH-JORGE1

```
ACCESS-SWITCH-JORGE1>
ACCESS-SWITCH-JORGE1>ena
ACCESS-SWITCH-JORGE1>enable
ACCESS-SWITCH-JORGE1#show spanning-tree vlan 90
VLAN0090
  Spanning tree enabled protocol rstp
  Root ID    Priority    24666
             Address     000A.4155.47B3
             Cost        23
             Port        23(FastEthernet0/23)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32858  (priority 32768 sys-id-ext 90)
             Address     0001.9624.54BA
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/23           Root FWD 19        128.23   P2p
Fa0/24           Altn BLK 19        128.24   P2p

ACCESS-SWITCH-JORGE1#
```

ACCESS-SWITCH-JORGE2

```
ACCESS-SWITCH-JORGE2#show spanning-tree vlan 90
VLAN0090
  Spanning tree enabled protocol rstp
  Root ID    Priority    24666
             Address     000A.4155.47B3
             Cost        23
             Port        24(FastEthernet0/24)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32858  (priority 32768 sys-id-ext 90)
             Address     00E0.8F76.A3DB
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/23           Altn BLK 19        128.23   P2p
Fa0/24           Root FWD 19        128.24   P2p

ACCESS-SWITCH-JORGE2#
```

DIST-SWITCH-JORGE1

```
DIST-SWITCH-JORGE1>ena
DIST-SWITCH-JORGE1>enable
DIST-SWITCH-JORGE1#show spanning-tree vlan 90
VLAN0090
  Spanning tree enabled protocol ieee
  Root ID    Priority    24666
             Address     000A.4155.47B3
             Cost        4
             Port        24(GigabitEthernet1/0/24)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32858  (priority 32768 sys-id-ext 90)
             Address     00D0.BA15.B675
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Gi1/0/1          Desg FWD 19        128.1    P2p
Gi1/0/2          Desg FWD 19        128.2    P2p
Gi1/0/10         Desg FWD 4         128.10   P2p
Gi1/0/11         Desg FWD 4         128.11   P2p
Gi1/0/23         Altn BLK 4         128.23   P2p
Gi1/0/24         Root FWD 4         128.24   P2p

DIST-SWITCH-JORGE1#
```

DIST-SWITCH-JORGE2

```
[OK]
DIST-SWITCH-JORGE2#
DIST-SWITCH-JORGE2#show spanning-tree vlan 90
VLAN0090
  Spanning tree enabled protocol rstp
  Root ID    Priority    24666
             Address     000A.4155.47B3
             Cost        8
             Port        24(GigabitEthernet1/0/24)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32858  (priority 32768 sys-id-ext 90)
             Address     0001.64D5.7288
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Gi1/0/2          Desg FWD 19        128.2    P2p
Gi1/0/1          Desg FWD 19        128.1    P2p
Gi1/0/10         Altn BLK 4         128.10   P2p
Gi1/0/11         Altn BLK 4         128.11   P2p
Gi1/0/23         Altn BLK 4         128.23   P2p
Gi1/0/24         Root FWD 4         128.24   P2p

DIST-SWITCH-JORGE2#
```

CORE-SWITCH-JORGE1

```
CORE-SWITCH-JORGE1>
CORE-SWITCH-JORGE1>enab
CORE-SWITCH-JORGE1>enable
CORE-SWITCH-JORGE1#show spanning-tree vlan 90
VLAN0090
  Spanning tree enabled protocol rstp
  Root ID    Priority    24666
             Address     000A.4155.47B3
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24666  (priority 24576 sys-id-ext 90)
             Address     000A.4155.47B3
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Gi1/0/1          Desg FWD 4         128.1    P2p
Gi1/0/2          Desg FWD 4         128.2    P2p
Gi1/0/20         Desg FWD 4         128.20   P2p
Gi1/0/21         Desg FWD 4         128.21   P2p

CORE-SWITCH-JORGE1#show spanning-tree vlan 180
VLAN0180
  Spanning tree enabled protocol rstp
  Root ID    Priority    24756
             Address     000A.4155.47B3
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24756  (priority 24576 sys-id-ext 180)
             Address     000A.4155.47B3
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Gi1/0/1          Desg FWD 4         128.1    P2p
Gi1/0/2          Desg FWD 4         128.2    P2p
```

CORE-SWITCH-JORGE2

```
CORE-SWITCH-JORGE2#
CORE-SWITCH-JORGE2#show spanning-tree vlan 90
VLAN0090
  Spanning tree enabled protocol rstp
  Root ID    Priority    24666
             Address     000A.4155.47B3
             Cost        4
             Port        20(GigabitEthernet1/0/20)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28762  (priority 28672 sys-id-ext 90)
             Address     00E0.A3AE.D935
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Gi1/0/1          Desg FWD 4         128.1    P2p
Gi1/0/2          Desg FWD 4         128.2    P2p
Gi1/0/21         Altn BLK 4         128.21   P2p
Gi1/0/20         Root FWD 4         128.20   P2p

CORE-SWITCH-JORGE2#show spanning-tree vlan 180
VLAN0180
  Spanning tree enabled protocol rstp
  Root ID    Priority    24756
             Address     000A.4155.47B3
             Cost        4
             Port        20(GigabitEthernet1/0/20)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28852  (priority 28672 sys-id-ext 180)
             Address     00E0.A3AE.D935
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Gi1/0/1          Desg FWD 4         128.1    P2p
Gi1/0/2          Desg FWD 4         128.2    P2p
```

2.3 Glosario rápido (STP)

- **FWD**: el puerto reenvía tráfico.
- **BLK**: el puerto está bloqueado por STP.
- **Root**: puerto con el mejor camino hacia el Root Bridge.
- **Desg (Designated)**: puerto designado como mejor salida en un segmento.
- **Altn (Alternate)**: camino alterno (normalmente bloqueado).

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3. Análisis por switch (por VLAN)

> Nota: Si en una captura aparece `LRN` (Learning), espera unos segundos y vuelve a ejecutar el comando hasta que cambie a `FWD/BLK`.

3.1 CORE-SWITCH-JORGE1 (Root Bridge)

VLAN 90, 180, 270 y 360

- **Modo:** Rapid-PVST (RSTP).
- **Rol del switch:** Root Bridge.
- **Puertos hacia otros switches (Gi1/0/1, Gi1/0/2, Gi1/0/20, Gi1/0/21):**
  - **Desg / FWD** (no hay puertos bloqueados en este switch).

3.2 CORE-SWITCH-JORGE2 (secundario)

VLAN 90

- **No es Root Bridge** (Root: CORE-SWITCH-JORGE1).
- **Gi1/0/20:** Root / FWD (camino principal al Root Bridge).
- **Gi1/0/21:** Altn / **BLK** (camino redundante bloqueado).
- **Gi1/0/1 y Gi1/0/2:** Desg / FWD.

VLAN 180, 270 y 360

Puerto	Estado	Rol
Gi1/0/1	FWD	Desg
Gi1/0/2	FWD	Desg
Gi1/0/20	FWD	Root
Gi1/0/21	**BLK**	**Altn**

3.3 DIST-SWITCH-JORGE1

VLAN 90, 180, 270 y 360

- **Root Bridge:** CORE-SWITCH-JORGE1.
- **Gi1/0/24:** Root / FWD (camino principal).
- **Gi1/0/23:** Altn / **BLK** (camino redundante).
- **Gi1/0/1, Gi1/0/2, Gi1/0/10, Gi1/0/11:** Desg / FWD.

3.4 DIST-SWITCH-JORGE2

VLAN 90

- **Gi1/0/24:** Root / FWD.
- **Gi1/0/10, Gi1/0/11, Gi1/0/23:** Altn / **BLK**.
- **Gi1/0/1 y Gi1/0/2:** Desg / FWD.

3.5 ACCESS-SWITCH-JORGE1

VLAN 90

- **Fa0/23:** Root / FWD.
- **Fa0/24:** Altn / **BLK**.
- **Fa0/1:** Desg / FWD (host).

VLAN 180

- **Fa0/2:** Desg / FWD.
- **Fa0/23:** Root / FWD.
- **Fa0/24:** Altn / **BLK**.

VLAN 270 y 360

- **Fa0/23:** Root / FWD.
- **Fa0/24:** Altn / **BLK**.

3.6 ACCESS-SWITCH-JORGE2

VLAN 90

- **Fa0/24:** Root / FWD.
- **Fa0/23:** Altn / **BLK**.

VLAN 180

- **Fa0/24:** Root / FWD.
- **Fa0/23:** Altn / **BLK**.

VLAN 270

- **Fa0/24:** Root / FWD.

- **Fa0/23:** Altn / **BLK**.
- **Fa0/1:** Desg / FWD.

VLAN 360

- **Fa0/24:** Root / FWD.
- **Fa0/23:** Altn / **BLK**.
- **Fa0/2:** Desg / FWD.

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4. Tabla

Switch	Interfaz	STP mode / prioridad	Estado	Rol
CORE1	Gi1/0/1	RSTP - 24666	FWD	Desg
	Gi1/0/2	RSTP - 24666	FWD	Desg
	Gi1/0/20	RSTP - 24666	FWD	Desg
	Gi1/0/21	RSTP - 24666	FWD	Desg
CORE2	Gi1/0/1	RSTP - 28762	FWD	Desg
	Gi1/0/2	RSTP - 28762	FWD	Desg
	Gi1/0/20	RSTP - 28762	FWD	Root
	Gi1/0/21	RSTP - 28762	BLK	Altn
DIST1	Gi1/0/1	RSTP - 32858	FWD	Desg
	Gi1/0/2	RSTP - 32858	FWD	Desg
	Gi1/0/23	RSTP - 32858	BLK	Altn
	Gi1/0/24	RSTP - 32858	FWD	Root
DIST2	Gi1/0/1	RSTP - 32858	FWD	Desg
	Gi1/0/2	RSTP - 32858	FWD	Desg
	Gi1/0/23	RSTP - 32858	BLK	Altn
	Gi1/0/24	RSTP - 32858	FWD	Root
ACCESS1	Fa0/23	RSTP - 32858	FWD	Root
	Fa0/24	RSTP - 32858	BLK	Altn
ACCESS2	Fa0/23	RSTP - 32858	BLK	Altn
	Fa0/24	RSTP - 32858	FWD	Root

5. Conclusión

La implementación de Rapid-PVST permitió mantener una topología redundante sin bucles de Capa 2. Mediante la elección del Root Bridge y la asignación de roles **Root/Designated/Alternate**, STP determinó qué enlaces permanecerían en **Forwarding** y cuáles debían bloquearse. Las verificaciones con `show spanning-tree` `vlan` confirmaron el funcionamiento del protocolo y la distribución de estados y roles en switches CORE, Distribution y Access.